PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Gold Digger AI Assistant

Version: 1.0

1. Project Overview (dear yam, aku dah tolong settle, tp kalau tak puas hati ekjah la, reply: awh sho shweet)

- **Problem Statement:** In Malaysia, gold prices fluctuate daily and sometimes hourly. Different shops offer different prices. Most buyers (from individuals to SMEs) make purchasing decisions based on emotion, intuition, or social media rumors rather than data. Without specialized financial skills, they struggle to interpret price charts, often buying at market peaks and missing optimal entry points.
- **Target Domain:** Financial Decision Intelligence and Personal Asset Optimization.
- **Proposed Solution Summary:** Gold Digger removes the guesswork by using the Z.AI GLM to transform raw market data into actionable strategies. It uses current prices, historical trends, and user contexts (budget and goals) to provide actionable recommendations and buying strategies rather than just displaying raw price charts.

2. Background & Business Objective

- **Background of the Problem:** Traditionally, buyers only have access to raw charts or shop prices. They lack a tool that cross-references their personal financial goals with market trends to give contextual advice. This gap prevents many from maximizing their savings.
- **Importance of Solving This Issue:** By accelerating price interpretation from hours of research to seconds, users can minimize risk and average purchase cost and maximize quantifiable savings by securing better entry prices.
- **Strategic Fit / Impact:** It aligns with the "Decision Intelligence" theme by demonstrating clear financial impact, easy demo-ability, and strong contextual reasoning.

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3. Product Purpose

3.1. Main Goal of the System: To provide an explainable decision support system that recommends optimal gold purchasing strategies based on trends, patterns, and personal goals.

3.2. Intended Users (Target Audience):

- First-time Gold Investors
- SME Jewelry Resellers
- Casual Savers (using gold as a hedge)

4. System Functionalities

4.1. Description: The Gold Digger is a financial decision intelligence engine that utilizes natural language processing to transform complex, fluctuating gold market data into personalized investment advice. It utilizes the Z.AI GLM to transform complex, fluctuating gold market data into personalized investment strategies. By moving beyond simple price charts, Gold Digger interprets market trends against specific user budgets and goals to provide explainable purchasing advice that removes the guesswork from gold investing.

4.2. Key Functionalities:

- **Trend Interpretation & Analysis:** Translates raw daily and hourly gold price fluctuations into actionable human-language insights, moving beyond simple price display.
- **Contextual Strategy Generation:** Cross-references personal financial goals (e.g., hedging, reselling) with real-time market data to recommend optimal entry points and buying strategies.
- **Risk Mitigation Alerts:** Identifies market peaks and "social media rumors" to prevent users from making emotional purchasing decisions

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- **Budget-Based Optimization:** Provides specific recommendations on how much and when to buy based on the user's defined budget.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Z.AI GLM. This model was chosen for its strong contextual reasoning capabilities and its ability for Gold Digger to transform unstructured market data into "Decision Intelligence".

4.3.2. **Prompting Strategy:** Multi-step agentic prompting. This approach is necessary for Gold Digger to first parse market data, then weigh it against user constraints, and finally generate an explainable recommendation.

4.3.3. **Context & Input Handling**: The system accepts inputs up to 1000 tokens. If a user provides an overly complex history or narrative exceeding this limit, the system automatically truncates the input to prioritize the most recent and relevant data.

4.3.4. **Fallback & Failure Behavior:** To manage API limits or potential model failures, the system includes a 6-hour cache of general market analysis. If the live API fails or the model returns unusable data, Gold Digger can revert to this cached analysis to ensure continuous support.

5. User Stories & Use Cases

- **User Stories:** *"As a person with no financial background, I want Gold Digger to explain market trends in simple, everyday language so that I can understand when to buy without needing to learn how to read complex financial charts"*
- **Use Cases (Main Interactions):**

  "Explain It Like I'm 5" (Market Translation) Flow: This interaction ensures accessibility for non-experts by translating technical market charts and daily fluctuations into plain, human language to remove the barrier of needing

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"specialized financial skills". When a user inputs a simple query such as "Is today a good day to buy?", the system uses the Z.AI GLM to parse the intent, fetch real-time API data, and provide a clear recommendation, such as "Wait for a dip" or "Buy now" and accompanied by a jargon-free explanation. By prioritizing "Decision Intelligence" over complex data visualization, the system empowers first-time investors to understand market conditions in seconds rather than hours of research

Micro-Budget Optimization Flow: This flow makes gold investing accessible to everyone by calculating the most effective entry points based on a user's specific, even minimal, budget. A user can input a small amount, such as RM100, and the system will cross-reference this against current market trends and shop prices to identify the best time to purchase small grammages to maximize their value. This functionality ensures that casual savers and those with limited funds are not excluded from asset optimization, allowing them to build savings incrementally based on data-backed strategies rather than guesswork.

- **Use Case Diagram:** *(kalau rajin buat kot……..)*

6. Features Included (Scope Definition)

6.1 Core Decision Intelligence Features: AI Decision Recommendation Engine

The Gold Digger AI Assistant uses Z.AI's General Language Model (GLM) as the central reasoning engine to analyze live gold price data, historical trends, and user-defined constraints such as budget and investment goals. Instead of simply displaying price charts, the system transforms raw data into intelligent financial recommendations that help users decide the best time and strategy to purchase gold.

The system produces actionable outputs such as Buy Now, Wait for Price Dip, or Split Purchase Strategy, along with a clear explanation of the reasoning behind each

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recommendation. This ensures users receive meaningful decision intelligence rather than just data visualization.

6.2 Data Interpretation Features: Real-Time Gold Price Analysis

The system retrieves live gold price data from a gold price API and analyzes price movements in real time. The AI interprets fluctuations and identifies patterns such as increasing trends, decreasing trends, and volatility, allowing users to quickly understand market conditions.

By converting raw numerical data into human-readable insights, the system reduces the complexity of financial data and allows users to make faster and more informed decisions without needing specialized financial knowledge.

6. 3 Historical Trend Interpretation

The system analyzes historical gold price data to detect patterns and trends over time. This enables the AI to determine whether the market is experiencing upward growth, downward movement, or unstable fluctuations.

The interpreted trends are then translated into simple explanations such as whether the market is stable or volatile, helping users understand the context behind the recommendation and supporting better decision-making.

6.4 Context-Aware Reasoning Features: Goal-Based Strategy Optimization

The system adjusts recommendations according to the user's investment objective, such as long-term savings, short-term profit, or financial hedging. Different goals require different strategies, and the AI evaluates market conditions to produce goal-specific recommendations.

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This ensures the system provides relevant advice based on user intent, helping individuals and small businesses optimize their purchasing decisions according to their financial objectives.

6.5 User Interface Features: Chat Interface

The system provides a conversational interface that allows users to interact with the AI using natural language. This simplifies the user experience and removes the need for technical knowledge or financial expertise.

Users can ask questions such as whether it is a good time to buy or how to invest based on their budget, and the AI responds with clear, explainable recommendations based on real-time market data and contextual reasoning.

7. Features Not Included (Scope Control)

- **Live Trading / Buy-Sell Execution System:** The system does not support real financial transactions or direct trading execution. Users cannot buy or sell gold through the platform. The AI only provides recommendations and insights for decision support, not actual execution of trades.

- **Shop Recommendation & Price Comparison Between Gold Stores:** The system does not include features that recommend where to physically buy gold or compare prices between different gold shops or dealers. This is intentionally excluded because real-world shop pricing varies frequently and requires verified commercial integrations, which is outside the scope of this AI assistant.

- **Full Backend Financial Infrastructure:** Advanced production-level systems such as microservices architecture, brokerage integration, or real-time trading pipelines are not implemented. The current system relies on external APIs for gold price data and AI-generated insights only.

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8. Assumptions & Constraints

- **LLM Cost Constraint:** A charge is made based on the amount of tokens processed and generated by the models. A single session might involve 1000 to 2000 tokens, which lead to an estimation of RM0.20 per session, following the current industry rates. To manage the costs, caches of general market analysis with a 6-hours window is included in the system's design.
- **Technical Constraints:** The prototype relies on a free-tier Gold Price API, which limits the system to 50 price refreshes per hour. Consequently, the information provided might become static if the limit is exceeded.
- **Performance Constraints:** With a token limitation, the system will automatically truncate the input to prioritize the most recent data if users provide an overly complex investment history or goal narrative exceeding tokens allocated.
- **User Input:** The model is unable to detect human-error. It assumes users input their budget accurately and define their risk or goals truthfully.

9. Risks & Questions Throughout Development

- **Market Volatility:** How can the system handle sudden price crashes during the 5-second generation window where historical trends do not apply?
- **Data Integrity:** What happens if the live Gold API fails during the final pitch?
- **User Reliance**: How do we ensure users understand that the purpose of the system is to act as a support tool rather than a guaranteed financial outcome?

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